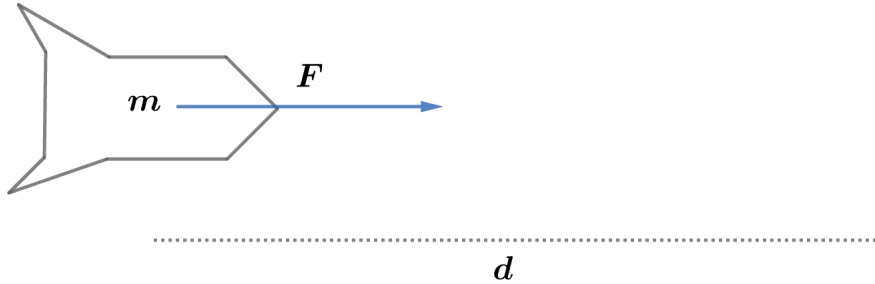


2007 F=ma Exam: Problem 22

Kevin S. Huang



Since a constant force F acts on the rocket over a distance d , the work done on the rocket is $W = Fd$. By the work-energy theorem, this is equal to the kinetic energy gained by the rocket,

$$W = Fd = \Delta K = \frac{p^2}{2m}$$
$$p = \sqrt{2Fdm}$$

If the same force F acts over the same distance d but the mass m is increased by a factor of 9, then the momentum p increases by a factor of 3 so the answer is D.